Registers

A/PSW	0x 00 54
BC	0x 00 00
DE	0x 00 00
HL	0x 00 00
SP	0x 47 FF
PC	0x 00 27

Flags

S	☐
Z	☑
AC	☑
P	☑
C	☐
V	☐
K	☐

Interrupts

TRAP	🟢
RST 5.5	🔴
RST 6.5	🔴
RST 7.5	🔴

```
1  LXI SP, 47FFH
2  LDA 4500H
3  MOV B, A
4  LDA 4501H
5  MOV C, A
6
7  MOV A, B
8  ANA C
9  STA 4502H
10
11 MOV A, B
12 ORA C
13 STA 4503H
14
15 MOV A, B
16 XRA C
17 STA 4504H
18
19 MOV A, B
20 ANI 0FH
21 STA 4505H
22
23 MOV A, B
24 ANI 0F0H
25 STA 4506H
26 HLT
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP, 47FFH
2	0x3	3A 00 45	LDA 4500H
3	0x6	47	MOV B, A
4	0x7	3A 01 45	LDA 4501H
5	0xA	4F	MOV C, A
6			
7	0xB	78	MOV A, B
8	0xC	A1	ANA C
9	0xD	32 02 45	STA 4502H
10			
11	0x10	78	MOV A, B
12	0x11	B1	ORA C
13	0x12	32 03 45	STA 4503H
14			
15	0x15	78	MOV A, B
16	0x16	A9	XRA C
17	0x17	32 04 45	STA 4504H
18			
19	0x1A	78	MOV A, B
20	0x1B	E6 0F	ANI 0FH
21	0x1D	32 05 45	STA 4505H
22			
23			
24			
25			
26			

Program ran successfully

Please check the left panel for updated state.

Registers

A/PSW	0x 00 54
BC	0x 00 00
DE	0x 00 00
HL	0x 00 00
SP	0x 47 FF
PC	0x 00 27

Flags

S	☐
Z	☑
AC	☑
P	☑
C	☐
V	☐
K	☐

Interrupts

TRAP	🟢
RST 5.5	🔴
RST 6.5	🔴
RST 7.5	🔴

```
1  LXI SP, 47FFH
2  LDA 4500H
3  MOV B, A
4  LDA 4501H
5  MOV C, A
6
7  MOV A, B
8  ANA C
9  STA 4502H
10
11 MOV A, B
12 ORA C
13 STA 4503H
14
15 MOV A, B
16 XRA C
17 STA 4504H
18
19 MOV A, B
20 ANI 0FH
21 STA 4505H
22
23 MOV A, B
24 ANI 0F0H
25 STA 4506H
26 HLT
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP, 47FFH
2	0x3	3A 00 45	LDA 4500H
3	0x6	47	MOV B, A
4	0x7	3A 01 45	LDA 4501H
5	0xA	4F	MOV C, A
6			
7	0xB	78	MOV A, B
8	0xC	A1	ANA C
9	0xD	32 02 45	STA 4502H
10			
11	0x10	78	MOV A, B
12	0x11	B1	ORA C
13	0x12	32 03 45	STA 4503H
14			
15	0x15	78	MOV A, B
16	0x16	A9	XRA C
17	0x17	32 04 45	STA 4504H
18			
19	0x1A	78	MOV A, B
20	0x1B	E6 0F	ANI 0FH
21	0x1D	32 05 45	STA 4505H
22			
23			
24			
25			
26			

Program ran successfully

Please check the left panel for updated state.

Registers

A/PSW	0x 00 54
BC	0x 00 00
DE	0x 00 00
HL	0x 00 00
SP	0x FF FF
PC	0x 00 1A

Flags

S	☐
Z	☑
AC	☑
P	☑
C	☐
V	☐
K	☐

Interrupts

TRAP	●
RST 5.5	●
RST 6.5	●
RST 7.5	●

```
1  LDA 4600H
2  MVI B, 08H
3  MVI C, 00H
4
5  COUNT: RLC
6  JNC SKIP
7  INR C
8
9  SKIP: DCR B
10 JNZ COUNT
11
12 MOV A, C
13 STA 4602H
14
15 ANI 01H
16 STA 4601H
17
18 HLT
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	3A 00 46	LDA 4600H
2	0x3	06 08	MVI B, 08H
3	0x5	0E 00	MVI C, 00H
4			
5	0x7	07	COUNT: RLC
6	0x8	D2 0C 00	JNC SKIP
7	0xB	0C	INR C
8			
9	0xC	05	SKIP: DCR B
10	0xD	C2 07 00	JNZ COUNT
11			
12	0x10	79	MOV A, C
13	0x11	32 02 46	STA 4602H
14			
15	0x14	E6 01	ANI 01H
16	0x16	32 01 46	STA 4601H
17			
18	0x19	76	HLT

Program ran successfully

Please check the left panel for updated state.

Registers

A/PSW	0x 00 54
BC	0x 00 00
DE	0x 00 00
HL	0x 00 00
SP	0x FF FF
PC	0x 00 2B

Flags

S	☐
Z	☑
AC	☑
P	☑
C	☐
V	☐
K	☐

Interrupts

TRAP	🟢
RST 5.5	🔴
RST 6.5	🔴
RST 7.5	🔴

```
1  LDA 4600H
2  ORI 24H
3  STA 4601H
4
5  LDA 4600H
6  ANI 7EH
7  STA 4602H
8
9  LDA 4600H
10 XRI 10H
11 STA 4603H
12
13 LDA 4600H
14 ANI 08H
15 JZ ZERO
16 MVI A, 01H
17 JMP STORE
18
19 ZERO: MVI A, 00H
20
21 STORE: STA 4604H
22 HLT
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	3A 00 46	LDA 4600H
2	0x3	F6 24	ORI 24H
3	0x5	32 01 46	STA 4601H
4			
5	0x8	3A 00 46	LDA 4600H
6	0xB	E6 7E	ANI 7EH
7	0xD	32 02 46	STA 4602H
8			
9	0x10	3A 00 46	LDA 4600H
10	0x13	EE 10	XRI 10H
11	0x15	32 03 46	STA 4603H
12			
13	0x18	3A 00 46	LDA 4600H
14	0x1B	E6 08	ANI 08H
15	0x1D	CA 25 00	JZ ZERO
16	0x20	3E 01	MVI A, 01H
17	0x22	C3 27 00	JMP STORE
18			
19	0x25	3E 00	ZERO: MVI A, 00H
20			
21	0x27	32 04 46	STORE: STA 4604H
22	0x2A	76	HLT

Program ran successfully

Please check the left panel for updated state.

Registers

A/PSW	0x 00 44
BC	0x 00 00
DE	0x 00 00
HL	0x 00 00
SP	0x FF FF
PC	0x 00 29

Flags

S	☐
Z	☑
AC	☐
P	☑
C	☐
V	☐
K	☐

Interrupts

TRAP	●
RST 5.5	●
RST 6.5	●
RST 7.5	●

```
1  LDA 4600H
2  RLC
3  STA 4601H
4
5  LDA 4600H
6  RRC
7  STA 4602H
8
9  LDA 4600H
10 ORA A
11 RAL
12 STA 4603H
13
14 LDA 4600H
15 ORA A
16 RAR
17 STA 4604H
18
19 LDA 4600H
20 MOV B, A
21 RAL
22 MOV A, B
23 RAR
24 STA 4605H
25
26 HLT
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	3A 00 46	LDA 4600H
2	0x3	07	RLC
3	0x4	32 01 46	STA 4601H
4			
5	0x7	3A 00 46	LDA 4600H
6	0xA	0F	RRC
7	0xB	32 02 46	STA 4602H
8			
9	0xE	3A 00 46	LDA 4600H
10	0x11	B7	ORA A
11	0x12	17	RAL
12	0x13	32 03 46	STA 4603H
13			
14	0x16	3A 00 46	LDA 4600H
15	0x19	B7	ORA A
16	0x1A	1F	RAR
17	0x1B	32 04 46	STA 4604H
18			
19	0x1E	3A 00 46	LDA 4600H
20	0x21	47	MOV B, A
21	0x22	17	RAL
22	0x23	78	MOV A, B
23			
24			
25			
26			

Program ran successfully

Please check the left panel for updated state.